ARtPUT: Autonomous Rover to Pick Up Trash

Progress Report 5: Progressing with the Machine Learning Algorithm

Bokre Samson, Dylan Shattan, Brendan Amoah-Daniels

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Materials and Methods

Materials:

- DFRobot Devastator Tank Mobile Robot Platform serving as the tracked mobile base
- Raspberry Pi 4 Model B as the primary onboard computer for vision, navigation, and logging
- Arduino UNO for low-level motor, servo, and sensor control
- USB RGB camera with depth-sensing capability for object detection and grasp planning
- Ultrasonic sensors for short-range obstacle detection
- NEO-6M GPS module for position logging and route tracking
- Digital compass (magnetometer) for heading correction
- Infrared sensor for bin fill-level detection
- Two-finger robotic gripper actuated by a servo motor
- 12V DC suction pump with relay module
- Suction-assisted collection bin with approximately 500–1000 mL capacity
- L298N motor drivers for drive motors
- Buck converters for voltage regulation
- Two 11.1 V Li-Po batteries to power onboard systems
- Wireless telemetry and onboard data logging storage
- Cones, measuring tape, stopwatch, and field notebook for controlled outdoor testing

Procedure:

1. Dataset & training: To train an object-detection model (YOLO/SSD-style), gather and label a significant number of photos of the target litter types in the target park. Use artificial occlusions to enhance.
2. Software stack: Put in place a perception pipeline (object detection on RGB-D + depth clustering), SLAM-based localization (Lidar + odometry), and a manipulation controller that translates detected object pose to a grasp technique. We'll test two modes of perception manipulation: Vision-only bounding-box grasping (A); vision plus depth-point-cloud grasp planning (B).
3. Three 10 m × 10 m test plots in a public park with low pedestrian traffic are chosen for the test. A fixed number (N = 20) of different litter items are placed in each plot at random locations (pre-approved with park authority). Mark the start and goal points.
4. Trials: For each mode (A, B) run 10 trials per plot (total 60 runs), each trial starting from the same location, time-limited to 15 minutes or until bin full. Rotate item arrangements between trials. Record video, sensor logs, and manual observer notes.
5. Baseline: Include 10 human-assisted pickup runs (human uses same route, picks items by hand) for performance comparison.

Data Collection Methods:

- Automated logs: timestamps of detections, attempted grasps, success/failure flags, battery consumption, distance travelled.
- Manual annotations: items missed, misclassifications, pedestrian interactions, environmental notes (wetness, wind).

- Post-run bin content check to count unique items collected.

Safety Concerns:

- The use of electronics and batteries in our project presents a risk of burns and electrical shock.
- The gears in our system may catch onto our limbs and clothing, potentially crushing our limbs.
- The rover may move erratically and hit someone, causing a harmful impact.
- We may drop heavy objects on our feet.

Risk Mitigation:

- We will always wear safety glasses, gloves, and proper footwear while working on our project.
- We will not wear baggy clothes that could potentially get caught in gears.
- There will always be a supervising adult while we are working on our project.
- The rover will always be fully powered off when working on it.

Data Analysis

Since the last progress report, our group has continued working toward the goals that were previously established. These goals included determining how sensors and controllers would eventually be mounted onto the rover base, continuing development of the machine-learning algorithm, and confirming that the necessary project materials were either delivered or on the way.

In terms of the rover itself, the base that was assembled during the previous stage remains completed and ready for future integration. During this period, we continued planning how sensors, controllers, and other electronic components will eventually be mounted onto the rover. While full hardware integration has not yet begun, this planning stage has helped us determine how the system will be organized when installation starts.

The largest area of progress since the previous report has been the development of the machine-learning detection system. As mentioned previously, we began refurbishing an algorithm originally created by Bokre during a past project that classified different species of ticks. While that algorithm served as a useful starting point conceptually, the system used for this project has now been rebuilt using a modern object-detection model.

The current detection model is based on the YOLOv11 (You Only Look Once) architecture and was trained using tools provided by Roboflow. Roboflow provides curated datasets and an online training environment that allows object-detection models to be trained and evaluated efficiently.

For the initial stage of training, we focused on a single object category: plastic bottles. Using a dataset available through Roboflow, we trained the model to recognize and detect bottles within almost 4,000 images. Starting with one category allows us to verify that the model works correctly before expanding the system to detect additional types of trash.

After training the model, we evaluated its performance using the training metrics provided by Roboflow. The most recent training runs produced very strong results, with accuracy metrics reaching the 90% range. These results demonstrate that the model is already performing well at detecting bottles in images and provides a strong foundation for expanding the system to recognize other types of litter in the future.

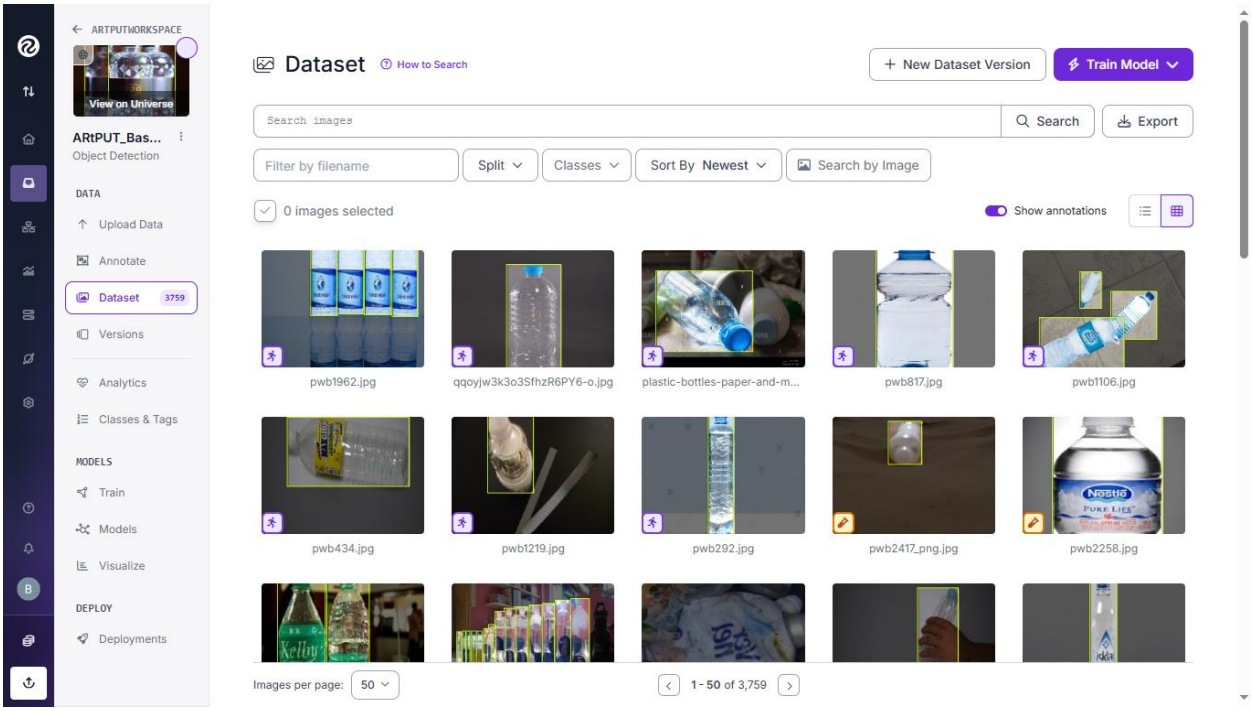


Figure 1: Screenshot showing the dataset used for training within Roboflow.

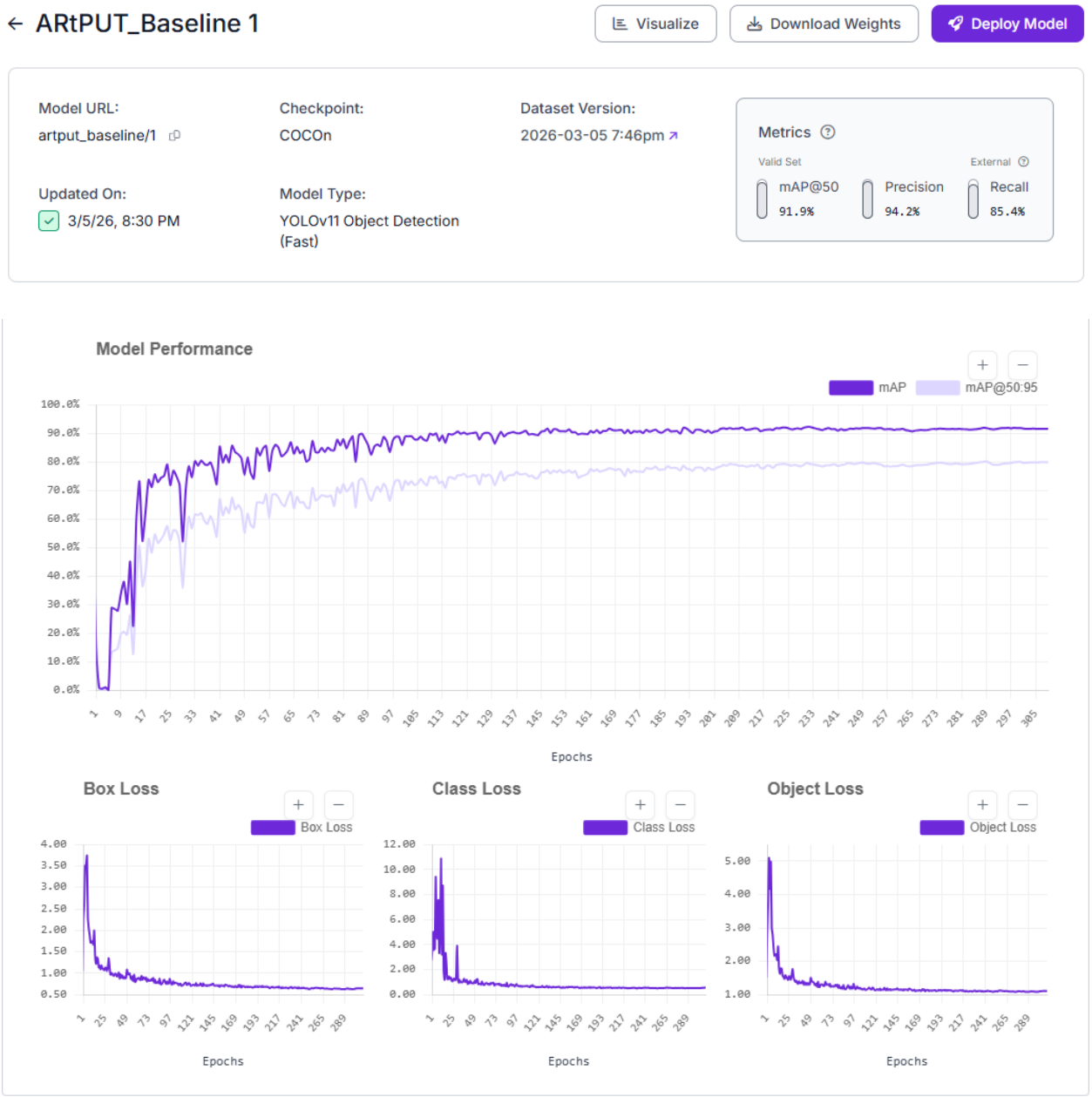
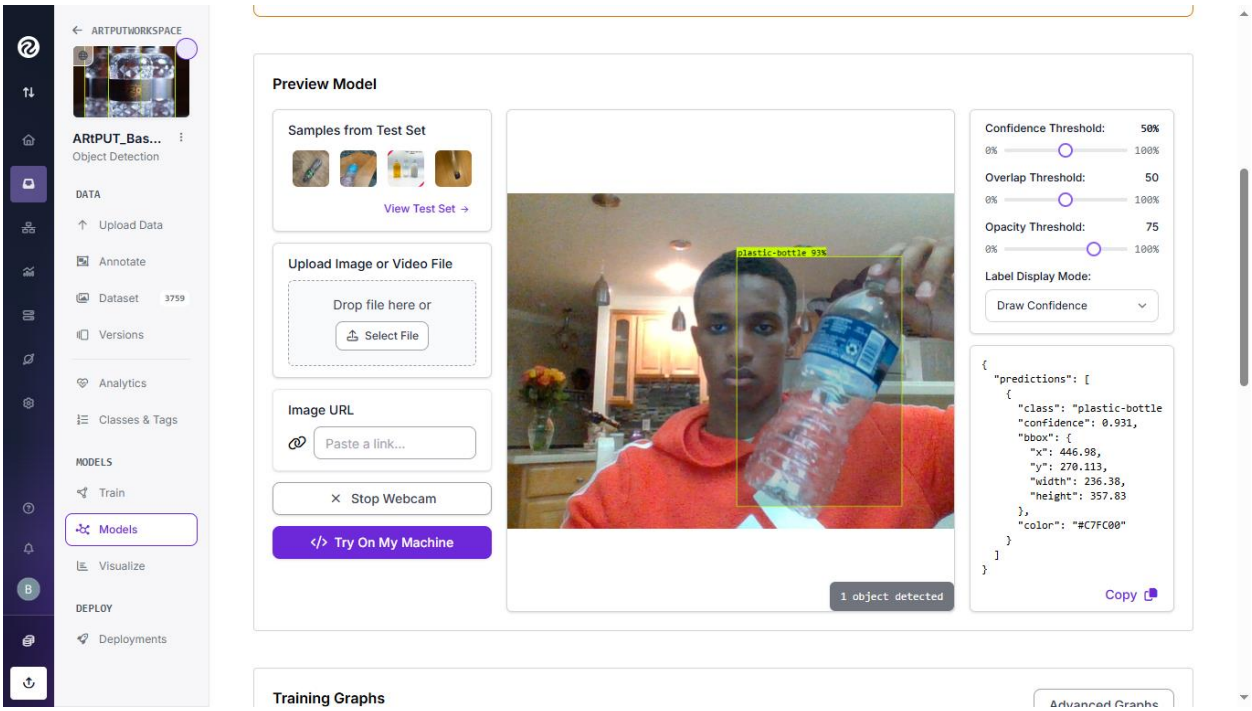


Figure 2-3: Screenshots displaying the training results and performance metrics of the YOLOv11 model.



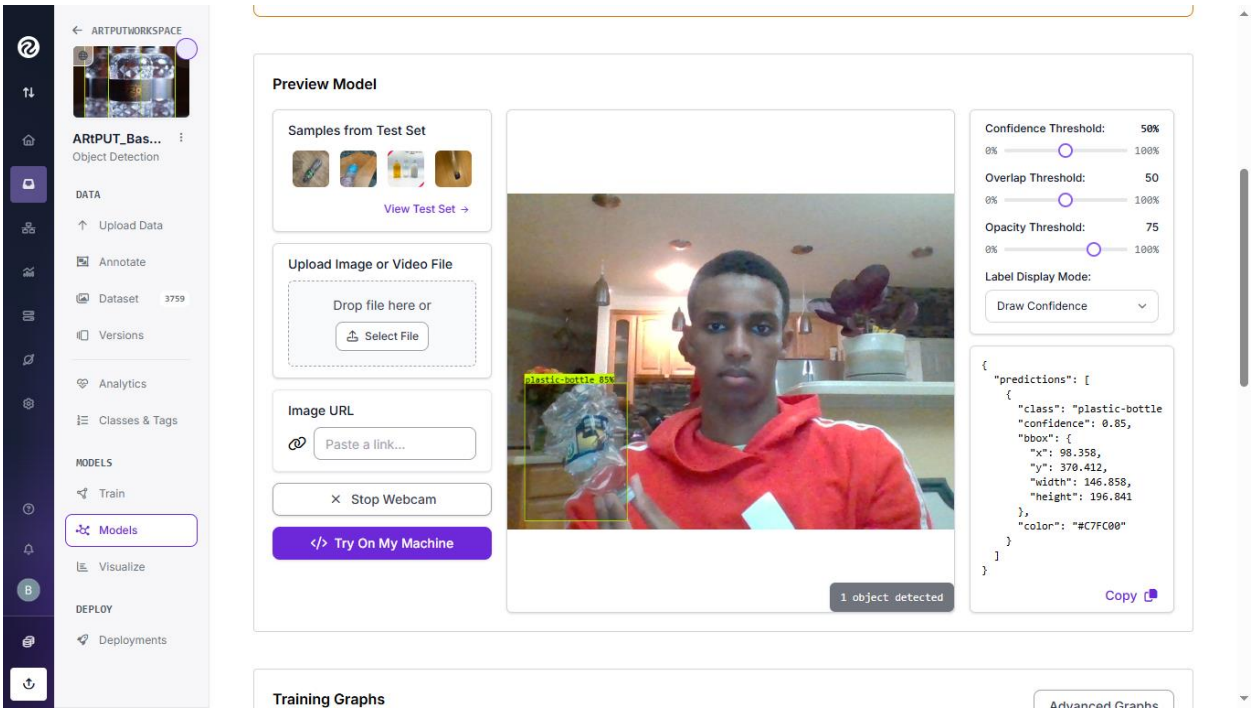


Figure 4-5: Screenshots showing example detections produced by the trained model.

Looking ahead, our goals for the next progress report include expanding the machine-learning model to detect additional types of trash objects, continuing to improve detection

accuracy through further training, and preparing the detection system for eventual integration with the rover's camera hardware.

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